DIPLOMA PROGRAMME PRACTICE · HL · SET B

Chemistry / Paper 2

Question booklet · 90 marks · 150 minutes

Answer every part. Show working, units and suitable precision. Use a calculator and a clean Chemistry data booklet. Use separate answer paper for additional working and diagrams.

QUESTION 1 [15]

Locating a period-3 element

Element X belongs to period 3. Its first four successive ionization energies are 578, 1817, 2745 and 11578 kJ mol^{-1}. In a separate experiment an isolated X atom absorbs a photon just sufficient for its first ionization. Use $h=6.63\times10^{-34}$ J s, $c=3.00\times10^{8}$ m s^{-1} and $N_A=6.02\times10^{23}$ mol^{-1}.

(a) Identify X and use the pattern, rather than a lookup, to explain your assignment. **[3]**

(b) Give the full configuration of X, that of X^{3+}, and the type of orbital from which the fourth electron is removed. **[3]**

(c) Calculate the longest wavelength that can ionize the neutral atom in its ground state. **[3]**

(d) Compare ionizing a ground-state atom with producing an emission line between two bound states. Relate each to the convergence limit. **[3]**

(e) A 220 nm photon strikes ground-state X. Determine whether single-photon ionization is possible and explain why the atom need not absorb this photon even if its energy is below the threshold. **[3]**

QUESTION 2 [15]

Two fluorides in a sealed vessel

Molecular XeF_4 and SF_4 are compared in the gas phase. Xenon has eight valence electrons; sulfur has six. Treat all Xe–F and S–F bonds as single in the Lewis models.

(a) Deduce the number of electron domains, lone pairs and molecular geometry around Xe in XeF_4. **[3]**

(b) Deduce the geometry of SF_4 and explain the preferred position of its lone pair. **[3]**

(c) Compare the expected permanent dipole moments and distinguish bond from molecular polarity. **[3]**

(d) For the Lewis model of XeF_4, show the formal charge on Xe and one F. State why zero formal charge does not imply a nonpolar bond. **[3]**

(e) Replace one F of square planar XeF_4 by Cl without changing the arrangement. Predict whether the four ligand positions were equivalent before replacement, whether a dipole is expected after replacement, and why. **[3]**

QUESTION 3 [15]

Dissociation of nitrogen dioxide

For $2NO_2(g) \rightleftharpoons 2NO(g)+O_2(g)$, use $\Delta H^\circ=+114.2$ kJ and $\Delta S^\circ=+146.4$ J K^{-1} per reaction as written. Assume constant values over the range considered. For an ideal gas the activity is its partial pressure divided by 1.00 bar. Use $\Delta G=\Delta G^\circ+RT \ln Q$ and $R=8.31$ J K^{-1} mol^{-1}.

(a) Explain the sign of ΔS° and determine the standard-state crossover temperature. **[3]**

(b) Calculate ΔG° and K at 700 K. **[3]**

(c) At equilibrium $p(NO)/p(NO_2)=0.500$. Deduce the oxygen partial pressure, showing the expression for K. **[3]**

(d) Oxygen is selectively removed, briefly leaving $p(O_2)=0.100$ bar and the same $NO:NO_2$ pressure ratio. Calculate ΔG at 700 K and predict the net reaction direction. **[3]**

(e) Determine the heat released by forming 0.750 mol NO_2 via the reverse reaction. Explain why a catalyst does not change the heat per mole for fixed reactant and product states. **[3]**

QUESTION 4 [15]

A buffer with a limited reserve

At 298 K a solution is made from 40.00 cm³ of 0.2000 mol dm⁻³ methanoic acid and 30.00 cm³ of 0.2000 mol dm⁻³ sodium methanoate. $K_a = 1.80 \times 10^{-4}$ and $K_w = 1.00 \times 10^{-14}$. Assume additive volumes and ideal concentrations.

(a) Calculate the initial pH using the buffer approximation. **[3]**

(b) 10.00 cm³ of 0.1000 mol dm⁻³ HCl is added. Show the chemical change and calculate the new pH. **[3]**

(c) Determine the further volume of 0.1000 mol dm⁻³ HCl needed to consume the remaining analytical methanoate. Explain why the original buffer formula then fails. **[3]**

(d) At this point estimate the pH from the weak acid equilibrium, using a small-dissociation approximation and checking it afterwards. **[3]**

(e) Evaluate two claims: dilution by a factor of two doubles buffer capacity expressed per litre, and dilution substantially changes the original buffer pH. Distinguish total amount from concentration. **[3]**

QUESTION 5 [15]

Testing an elementary-step model

A gas reaction obeys $2NO+O_2 \rightarrow 2NO_2$. A proposed mechanism is $2NO \rightleftharpoons N_2O_2$ (fast pre-equilibrium), then $N_2O_2+O_2 \rightarrow 2NO_2$ (slow). At fixed initial concentrations the initial rate at 320 K is 2.50 times that at 300 K. Use $R=8.31\ J\ K^{-1}\ mol^{-1}$.

(a) Identify the intermediate and show that summing the forward steps gives the observed overall equation. **[3]**

(b) Derive the predicted rate law in terms of reactants only. **[3]**

(c) Predict the rate change if [NO] is halved and $[O_2]$ tripled at fixed temperature; state overall order and units for the observed k when concentration is in $mol\ dm^{-3}$. **[3]**

(d) Estimate the apparent activation energy from the temperature pair. **[3]**

(e) Assess whether agreement with this rate law proves the mechanism. Suggest additional evidence and explain why the apparent activation energy need not equal the slow step’s barrier. **[3]**

QUESTION 6 [15]

An ester and its hydrolysis products

A saturated acyclic ester has formula $C_5H_{10}O_2$. Its ^{1}H NMR spectrum has a 6H doublet, a 1H septet near 5.0 ppm and a 3H singlet near 2.0 ppm. A sample of 2.040 g reacts completely with aqueous NaOH; M(ester)=102.0 g mol^{-1}.

(a) Use the splitting and integration of the doublet and septet to identify a fragment and its location relative to oxygen. **[3]**

(b) Deduce the complete ester structure and explain the singlet, giving the systematic name. **[3]**

(c) Write the equation for alkaline hydrolysis and calculate the minimum volume of 0.2500 mol dm^{-3} NaOH needed. **[3]**

(d) An excess of aqueous acid is then added to the cooled mixture. Identify the organic acid formed, write the responsible ionic equation and identify a diagnostic IR change for the isolated dry acid compared with the original ester. **[3]**

(e) Predict the main product on controlled oxidation of the alcohol hydrolysis product and identify two expected changes in its ^{1}H NMR spectrum. **[3]**